Definition 2.1

There are two jobs. Of which one job can be completed in m ways, when it has completed in any one of these m ways, second job can be completed in n ways, then the two jobs in succession can be completed in $m \times n$ ways. This is called multiplication principle of counting.

Example 2.11

Find the number of 4 letter words, with or without meaning, which can be formed out of the letters of the word "NOTE", where the repetition of the letters is not allowed.

Solution

Let us allot four vacant places (boxes) to represent the four letters. N,O,T,E of the given word NOTE.



Fig. 2.2

The first place can be filled by any one of the 4 letters in 4 ways. Since repetition is not allowed, the number of ways of filling the second vacant place by any of the remaining 3 letters in 3 ways. In the similar way the third box can be filled in 2 ways and that of last box can be filled in 1 way. Hence by multiplication principle of counting, total number of words formed is $4 \times 3 \times 2 \times 1 = 24$ words.

Example 2.12

If each objective type questions having 4 choices, then find the total number of ways of answering the 4 questions.

Solution

Since each question can be answered in 4 ways, the total number of ways of answering 4 questions

$= 4 \times 4 \times 4 \times 4 = 256$ ways.

Example 2.13

How many 3 digits numbers can be formed if the repetition of digits is not allowed?

Solution

Let us allot 3 boxes to represent the digits of the 3 digit number

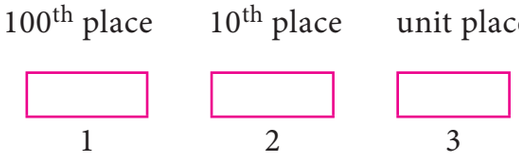


Fig. 2.3

Three digit number never begins with "0". $\therefore$ 100th place can be filled with any one of the digits from 1 to 9 in 9 ways.

Since repetition is not allowed, 10th place can be filled by the remaining 9 numbers (including 0) in 9 ways and the unit place can be filled by the remaining 8 numbers in 8 ways. Then by multiplication principle of counting, number of 3 digit numbers = $9 \times 9 \times 8 = 648$.

2.2.3 Addition principle of counting

Let us consider the following situation in a class, there are 10 boys and 8 girls. The class teacher wants to select either a boy or a girl to represent the class in a function. Our objective is to find the number of ways that the teacher can select the student. Here the teacher has to perform either of the following two jobs.

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